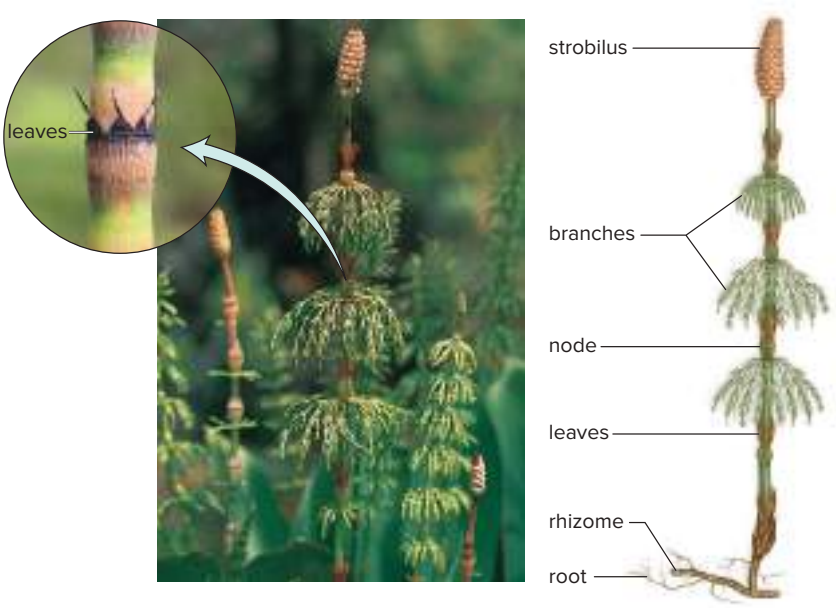


Figure 23.13 Horsetail, *Equisetum*. Whorls of branches and tiny leaves are at the nodes of the stem. Spore-producing sporangia are borne in strobili. (leaves): blickwinkel/Alamy Stock Photo; (plant): Robert Carr/Avalon

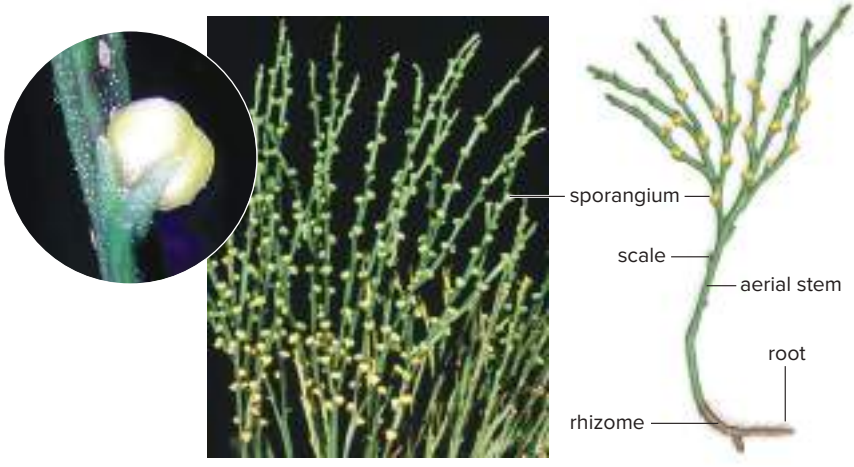


Figure 23.14 Whisk fern, *Psilotum*. *Psilotum* has no leaves—the branches carry on photosynthesis. The sporangia are yellow. (sporangium): Steven P. Lynch; (plant): Biophoto Associates/Science Source

on short side branches. The two or three species of *Tmesipteris* have appendages that some maintain are reduced megaphylls.

Ferns

Ferns are a widespread group of plants (~11,000 species) that are well-known for their attractiveness. Ferns are most abundant in warm, moist, tropical regions, but they can also be found in temperate regions and as far north as the Arctic Circle. Several species live in dry, rocky places, whereas others have adapted to an aquatic life. Ferns range in size from tiny aquatic species less than 1 cm in diameter to modern giant tropical tree ferns that exceed 20 m in height. Recall that unlike lycophytes, ferns have megaphylls, which are more commonly referred to as **fronds.** The leatherleaf fern (used in flower arrangements) has fronds that are broad, with subdivided leaflets; those of a tree fern can be about 1.4 meters long; and those of the hart's tongue fern are straplike and leathery (Fig. 23.15). Sporangia are often located in clusters, called *sori* (*sing., sorus*), on the undersides of the fronds, where they may be shielded by thin, protective structures called *indusia* (*sing., indusium*).

The life cycle of a typical temperate-zone fern, shown in Figure 23.16, applies in general to the other types of vascular seedless plants. The dominant sporophyte produces windblown spores by meiosis within sporangia. The windblown spores disperse, land, and germinate into the gametophyte, the generation that lacks vascular tissue. The separate, heart-shaped gametophyte produces flagellated sperm, which swim in a film of water from the antheridium to the egg within the archegonium, where fertilization occurs. Eventually, the gametophyte disappears and the sporophyte is independent. Both generations of a fern are considered to be independent of one another. Once established, some ferns, such as the bracken fern, can spread into drier areas, because their rhizomes, which grow horizontally in the soil, produce new plants.

Ferns have much economic value and are frequently used by florists in decorative bouquets and as ornamental plants in the home and garden. Wood from tropical tree ferns is often used as a building material, because it resists decay, particularly by termites. Ferns, especially the ostrich fern, are used as food—in the northeastern United States, many restaurants feature fiddleheads (that season's first



Figure 23.15 Diversity of ferns. Structural adaptations found in a variety of ferns. (leatherleaf fern): Gregory Preest/Alamy Stock Photo; (tree fern): ATOM WANG/Shutterstock; (hart's tongue fern): Nic Miller/Organics image library/Alamy Stock Photo

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